

ANDREW LIN

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EDUCATION

University of Waterloo

Bachelor of Computer Science, Co-op

Sep. 2022 – Apr. 2027

GPA: 3.93

- Relevant Courses: Data Structures, Algorithms, Operating Systems, Numerical Methods, Compilers

EXPERIENCE

Software Engineer Intern | Python, Django, PostgreSQL, AWS

Carta

Jan. 2026 – Present

Waterloo, ON

- Building automation for complex fund investment and distribution workflows, applying double-entry accounting and LLM-assisted entity classification to reduce manual intervention

Software Engineer Intern | C#, .NET, Terraform, GCP, Kubernetes

Geotab

Jan. 2025 – Apr. 2025

Waterloo, ON

- Developed and optimized telemetry pipelines handling real-time diagnostics for 200K vehicles across 20 partners
- Boosted data throughput by 15x for Ford vehicles by batching and grouping correlated vehicle events, eliminating Redis caching logic and reducing GCP Pub/Sub message streaming costs by \$30K annually
- Architected and deployed a cloud-native data ingestion and mapping pipeline for Kia vehicle integrations on Kubernetes, onboarding 1,000+ vehicles and reducing API calls by 30x using an event-driven task scheduler

Software Engineer Intern | Python, FastAPI, TypeScript, GCP

TD Bank

May 2024 – Aug. 2024

Toronto, ON

- Built and scaled a real-time speech translation app, securing \$100K in funding after VP demos and piloted in 12 branches
- Achieved a 6% increase in translation performance measured by string-based and ML-based metrics (e.g. BLEU, COMET) by evaluating and comparing the quality of four different translation models across 38 language pairs
- Implemented real-time translation mode for multi-language conversations by integrating streaming transcription using Whisper and incremental translation services in a FastAPI WebSocket server, enabling live speech-to-text translation

Software Engineer Intern | Python, GraphQL, TypeScript, AWS

Altis Labs

Jan. 2023 – Apr. 2023 & Sep. 2023 – Dec. 2023

Toronto, ON

- Engineered image annotation and data services for a clinical information platform, powering custom models assisting in segmentation and survival prediction on medical images using GraphQL, Python, and AWS
- Developed a synthetic data seeding tool to populate databases with 10+ GB of medical data and images for testing
- Overhauled patient-study labeling with new schemas and an Express.js validation service, boosting feature usage 300%

Research Assistant – Compilers | OCaml

University of Waterloo

May 2025 – Aug. 2025

Waterloo, ON

- Implemented code generation for four encodings of Bayesian networks using OCaml, enabling faster exact inference in a new probabilistic programming language under Prof. Yizhou Zhang (contributing to an [ICFP 2025 paper](#))
- Identified encoding trade-offs and optimized CPT representation in conditional branches, improving compile time by 20%

PROJECTS

CUDA-Enabled Deep Learning Library | C++, Python, CUDA, pybind11

- Implemented a C++ tensor library with strided storage and Python bindings, supporting core operations (matmul, ReLU, softmax, transpose) on CPU and naive CUDA, and built layer-wise backprop to achieve 92% accuracy on the MNIST dataset

Estimathon | TypeScript, Python, PostgreSQL, WebSockets

- Led team of 4 to design and launch a platform to host UW Data Science Club's team-based Estimathon contest every semester, supporting 100+ concurrent users and real-time updates using Next.js, PostgreSQL, and WebSockets

TECHNICAL SKILLS

Languages: Python, C++, C, C#, Java, TypeScript, OCaml, SQL, Bash

Libraries/Frameworks: FastAPI, PyTorch, pandas, numpy, CUDA, .NET, Spring Boot, React, Node.js

Developer Tools: Git, Docker, Kubernetes, Terraform, PostgreSQL, AWS (S3, RDS, Lambda), GCP (GKE, BigQuery, Pub/Sub)